8 A curve has parametric equations $x = \dfrac{t}{1+t^3}$, $y = \dfrac{t^2}{1+t^3}$, where $t \neq -1$.

(i) In this question you must show detailed reasoning.

Determine the gradient of the curve at the point where $t = 1$. **[5]**

(ii) Verify that the cartesian equation of the curve is $x^3 + y^3 = xy$. **[3]**